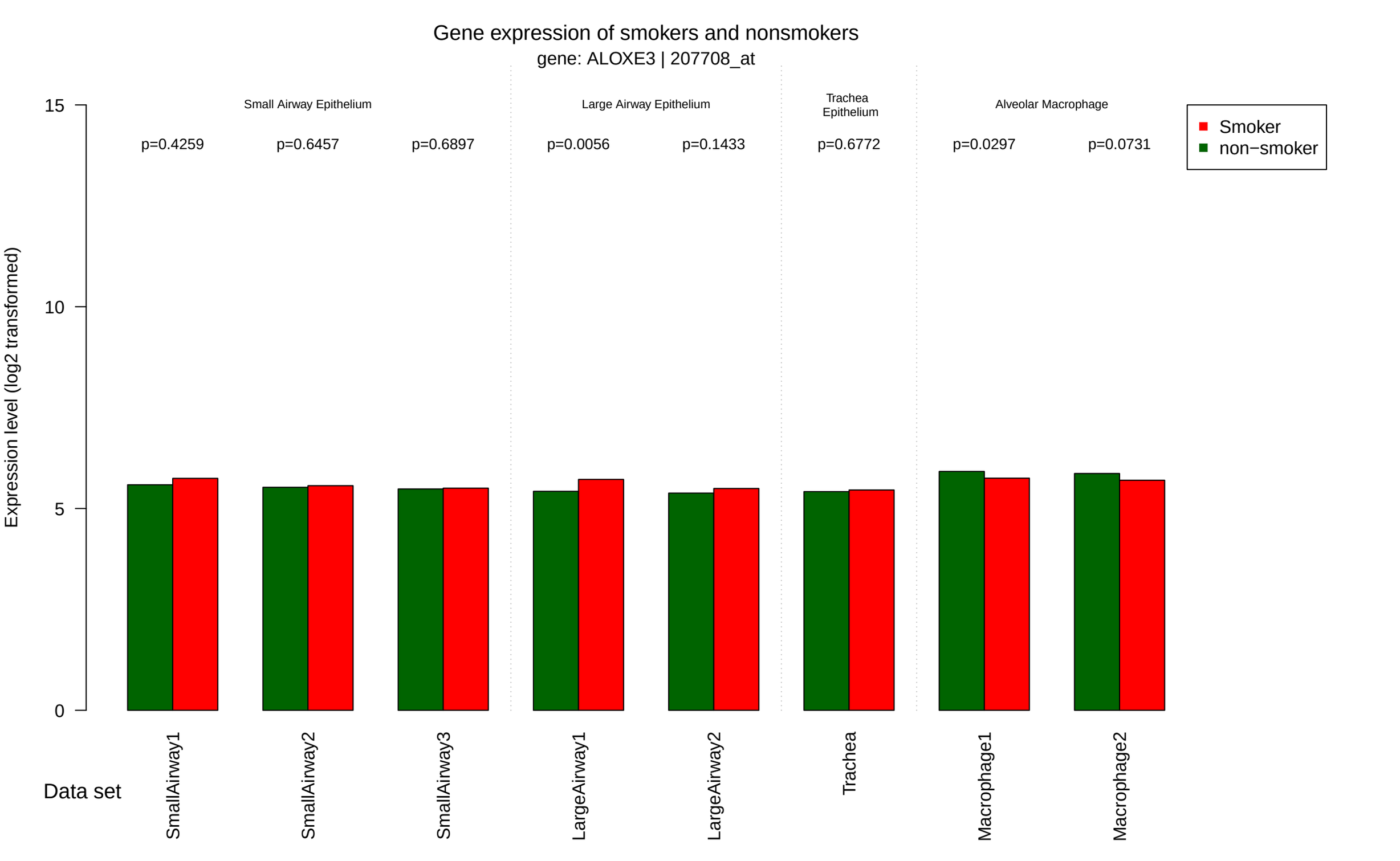
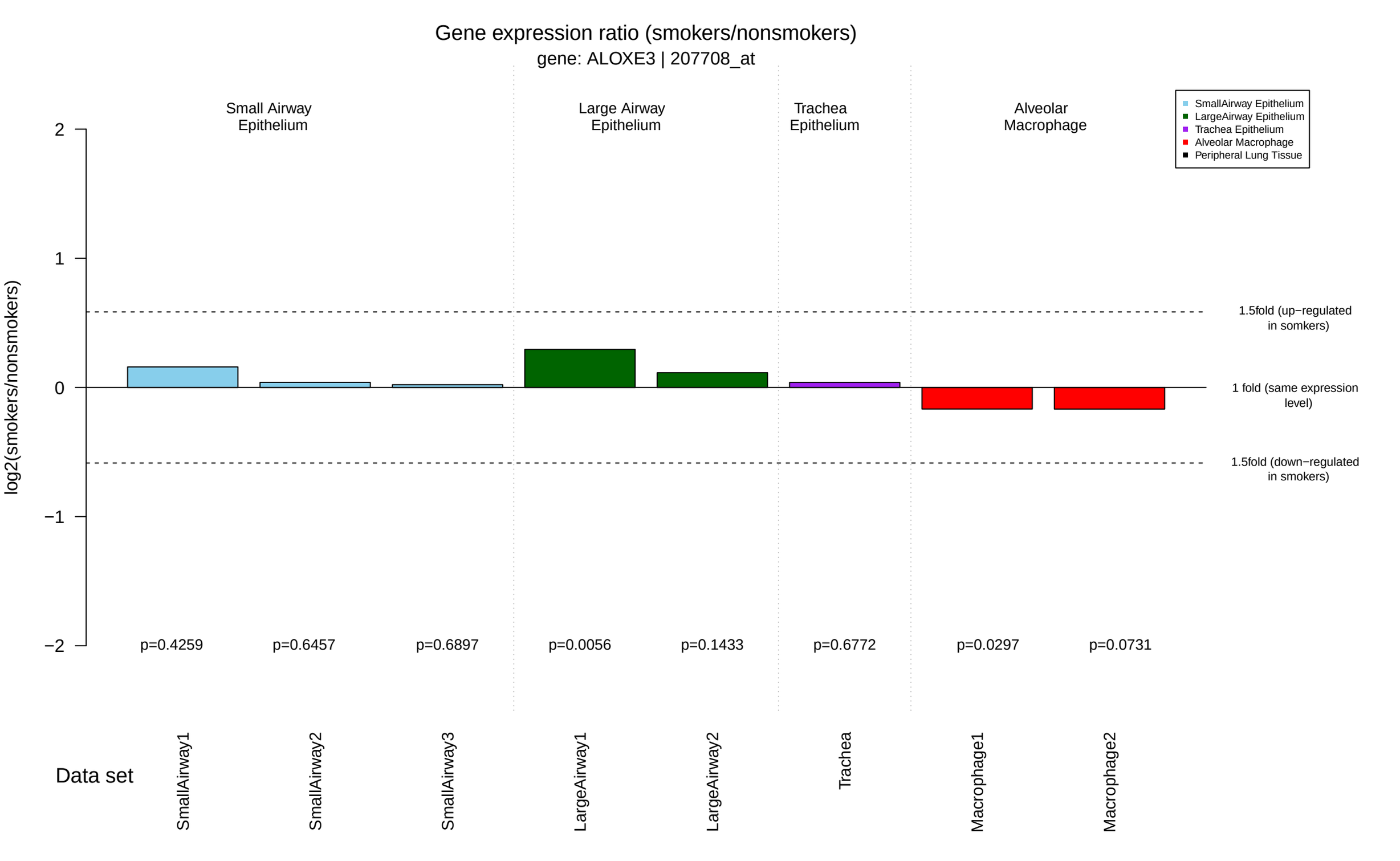
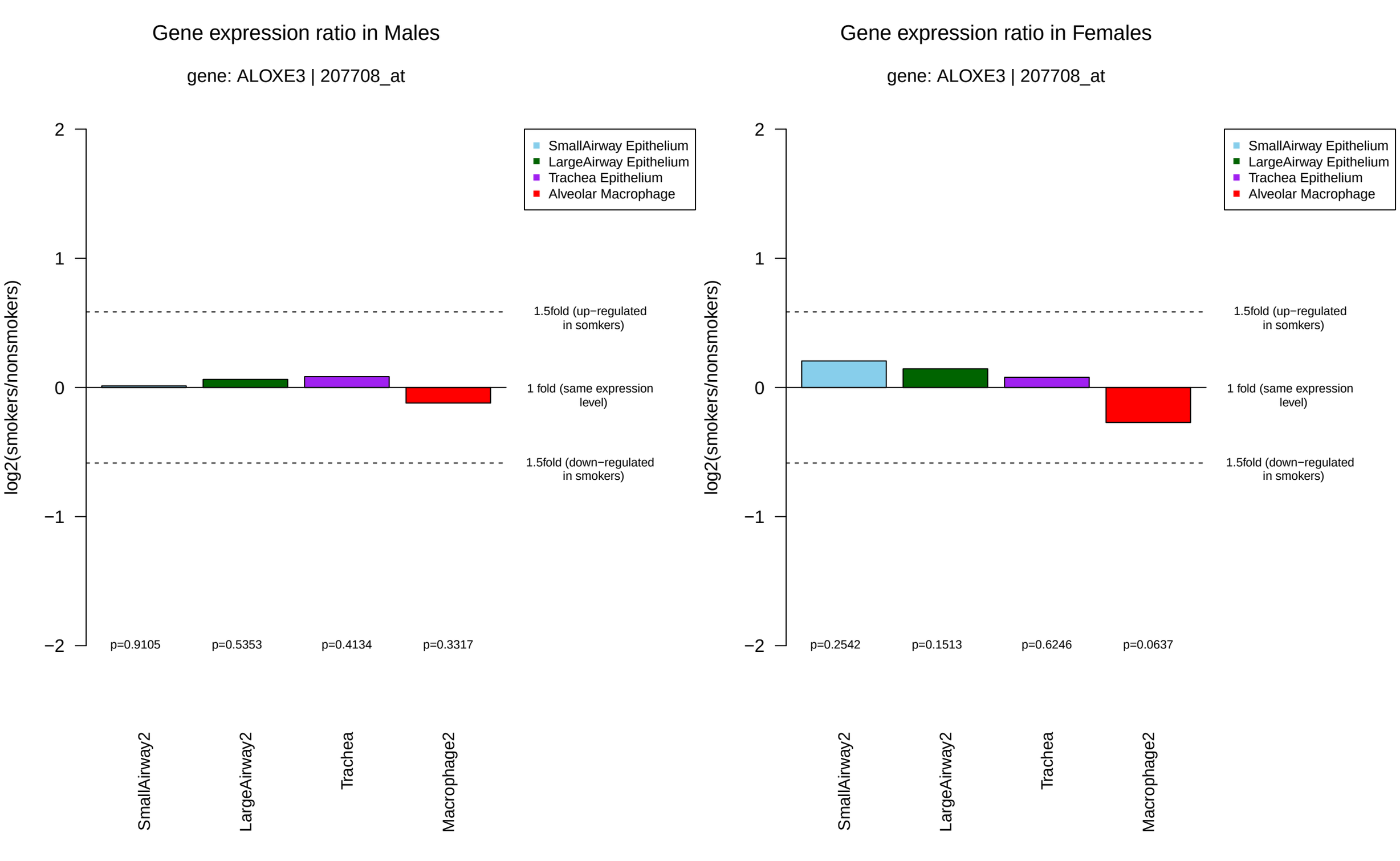
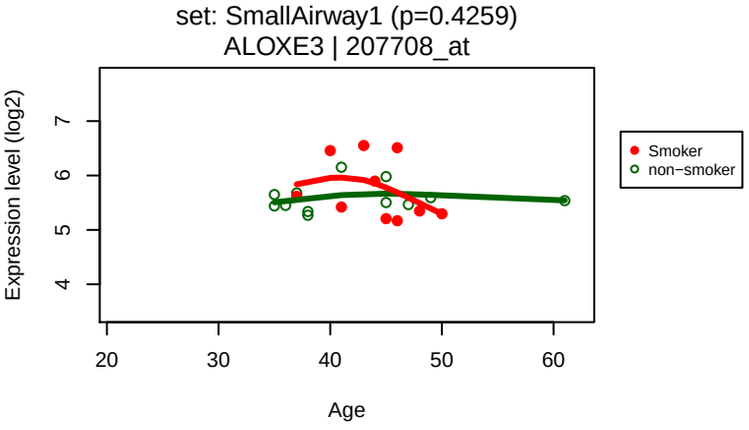
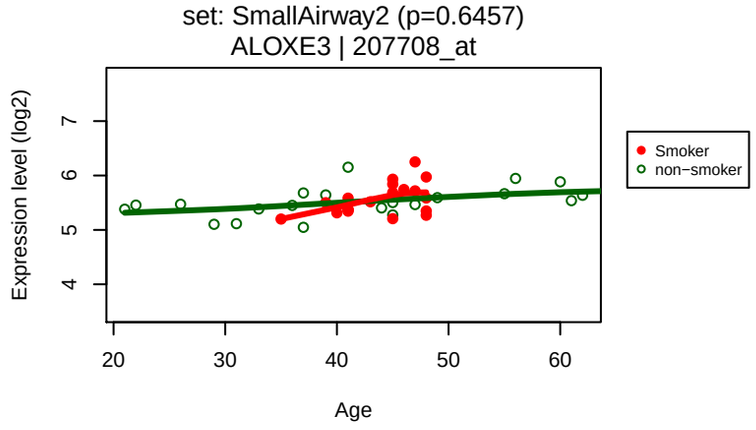
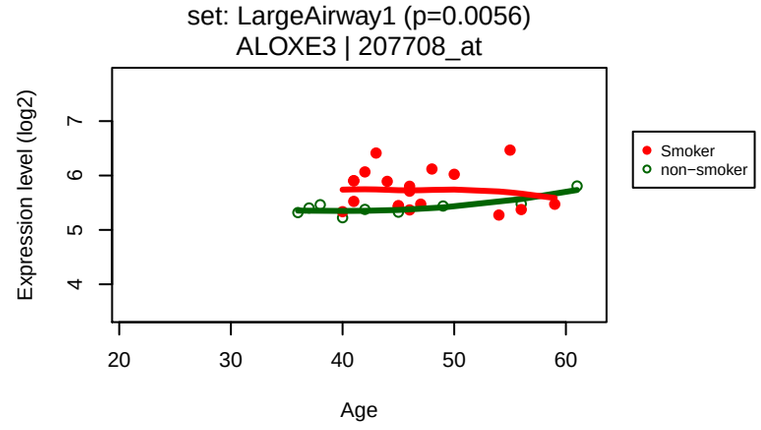

Gene expression of smokers and nonsmokers
gene: ALOXE3 | 207708_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4259
p=0.6457
p=0.6897
p=0.0056
p=0.1433
p=0.6772
p=0.0297
p=0.0731
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

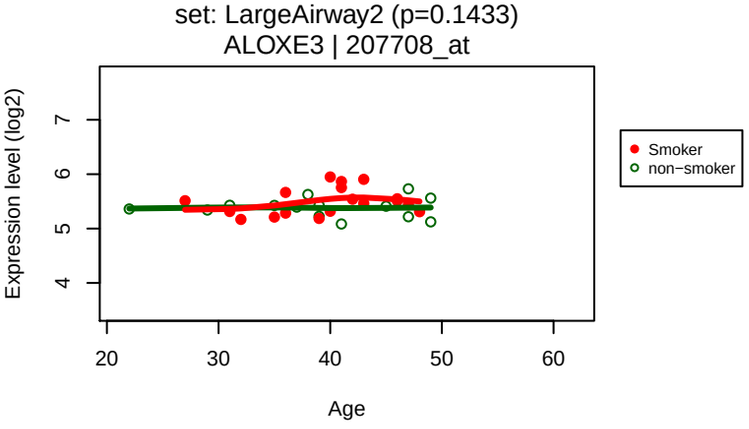

Gene expression ratio (smokers/nonsmokers)
gene: ALOXE3 | 207708_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4259
p=0.6457
p=0.6897
p=0.0056
p=0.1433
p=0.6772
p=0.0297
p=0.0731
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

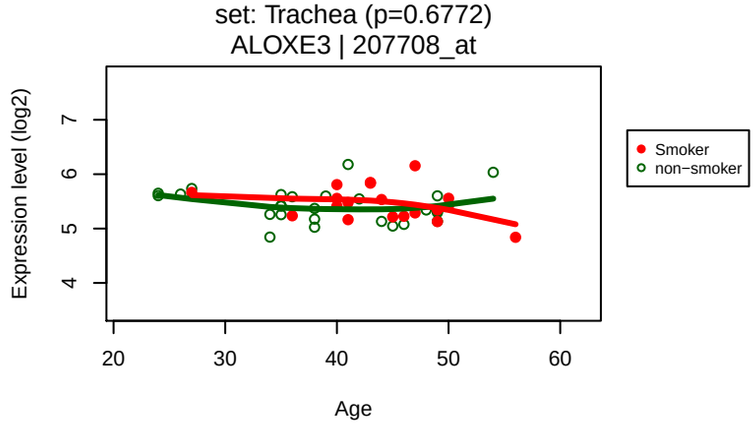

Gene expression ratio in Males
gene: ALOXE3 | 207708_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9105
p=0.5353
p=0.4134
p=0.3317
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOXE3 | 207708_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2542
p=0.1513
p=0.6246
p=0.0637
SmallAirway2
LargeAirway2
Trachea
Macrophage2

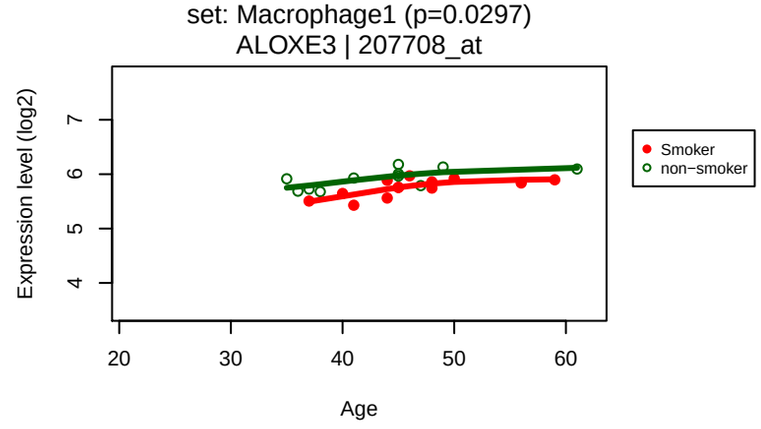

set: SmallAirway1 (p=0.4259)
ALOXE3 | 207708_at
Expression level (log2)
Age
Smoker
non−smoker

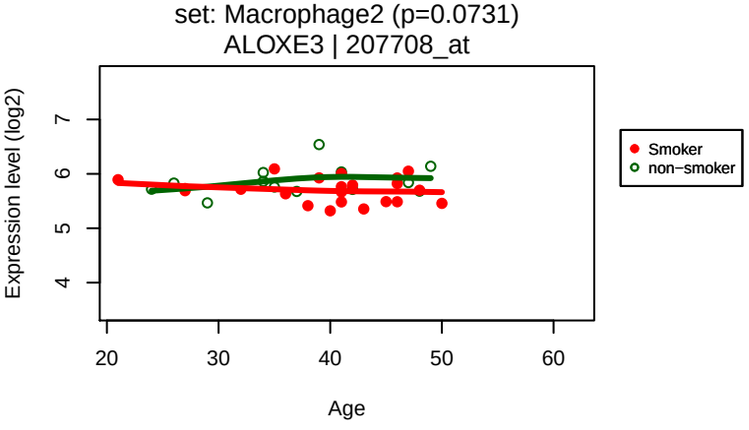

set: SmallAirway2 (p=0.6457)
ALOXE3 | 207708_at
Expression level (log2)
Age
Smoker
non−smoker

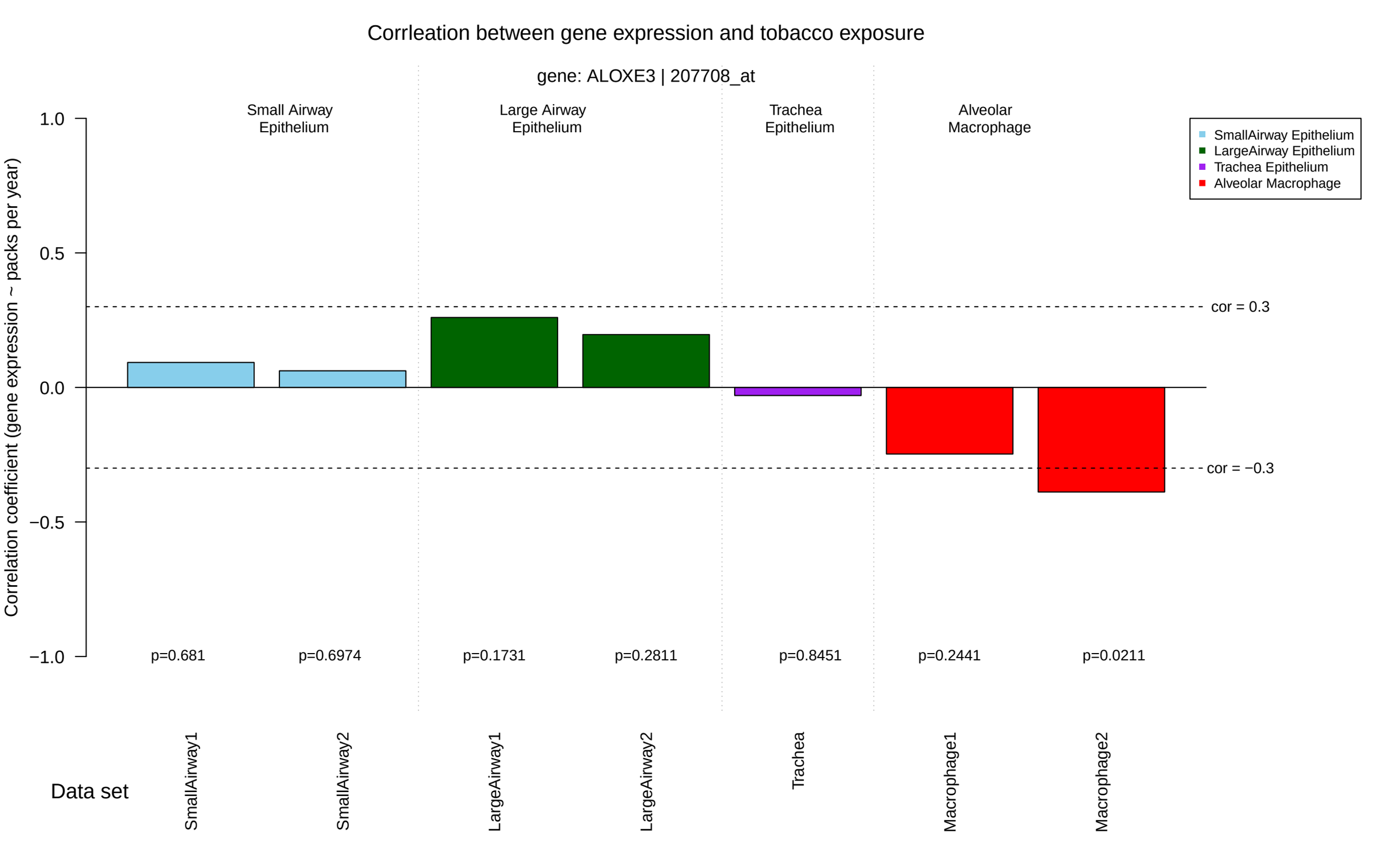

set: LargeAirway1 (p=0.0056)
ALOXE3 | 207708_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1433)
ALOXE3 | 207708_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6772)
ALOXE3 | 207708_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0297)
ALOXE3 | 207708_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0731)
ALOXE3 | 207708_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOXE3 | 207708_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.681
p=0.6974
p=0.1731
p=0.2811
p=0.8451
p=0.2441
p=0.0211
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2